Aya El Dassouki

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HIGHLIGHT OF QUALIFICATIONS

Fourth Year Computer Science student at York University with experience in **AI, machine learning,** and **data analysis**, including **LLMs, SQL, and Tableau**. Skilled in **software development, automation (Python, Docker),** and **technical leadership.**

EDUCATION Bachelor of Science in Computer Science (Honours) September 2022 – June 2026

Lassonde School of Engineering, York University ● **Activities:** York Engineering Competition **(1st Place Winner)**, AI in Healthcare Hackathon **(3rd Place Winner)**,

Women in Science and Engineering Club **Vice President**, Robotics Club Member ● **Courses:** Data Structures and Algorithms, Linux and shell scripting, Computer Architecture, Databases, Discrete Mathematics.

RELVENT WORK EXPERERINCE

Web developer Freelancer **February 2025-Present** *Space&Co* ● Developed and customized **a WordPress website** using **HTML, CSS, and automation tools**, improving **user experience** and **site functionality**.

- Researched and integrated **SEO best practices**, enhancing **website visibility and search ranking**.
- Collaborated closely with clients, gathering requirements, iterating on designs, and incorporating feedback to create an **engaging and professional brand presence online**.

Research Assistant **Sep 2024-Dec 2024**

Lassonde School of Engineering, Toronto ● Worked with **Professor Enas ALTarawneh** on **fine-tuning the Gemini model** for **multi-label emotion detection**, achieving a **77% improvement in F1-score** compared to non-fine-tuned models.

- Processed and analyzed **datasets**, optimizing preprocessing and prompt engineering, leading to a **20% increase in precision** for underperforming emotion categories.
- Conducted a **comparative evaluation** of **Llama 3, DistilBERT, and Gemini**, demonstrating Gemini's superior performance across all five emotion categories.

TECHNICAL PROJECTS

Customer Churn Prediction Analysis

- Developed a **machine learning pipeline** using Python, Pandas, and Scikit-learn to predict customer churn, **achieving 82.1% accuracy**. ● Performed **data preprocessing**, handling missing values and converting categorical features, leading to **improved model reliability**.
- Implemented **Logistic Regression**, balancing precision (86.2% for non-churn and 68.5% for churn) to **enhance customer retention insights**.

CNN Image Classifier

- Developed and trained a **Convolutional Neural Network (CNN)** using TensorFlow and Keras to classify images from the CIFAR-10 dataset**,** achieving **45% accuracy.**

- Implemented data preprocessing techniques such as normalization and data augmentation, leading to better model generalization.
- Containerized the model using **Docker,** streamlining deployment and enabling scalable and reproducible model execution across environments.

SKILLS

- **Languages & Tools:** Python, SQL, JavaScript, Tableau, Power BI, SQL Queries, Excel (Pivot Tables, Power Query, Macros)
- **Machine Learning & AI:** LLMs, TensorFlow, PyTorch, scikit-learn, Data Preprocessing, Model Deployment